

An Improved Mask-Guided Glaucoma Screening Method Based on Convolution Neural Network

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ABSTRACT

In recent years, computer intelligent technology has been widely used in clinical diagnosis. Glaucoma is concealed and difficult to cure, so the early diagnosis is vital important for prevention. The existing automatic screening methods still have many problems in practical applications, such as excessive dependence on the segmentation accuracy of fundus images, rely on a large amount of data, poor interpretability, and so on. To solve these limitations, we propose a mask-guided glaucoma screening method. First, we segment the color fundus image to obtain the corresponding mask image. Then, the mask image and the fundus image are fused and input into our proposed classification network MG-Efficient Net for glaucoma prediction. In particular, mask image is not directly used for glaucoma classification, so the segmentation accuracy has little influence on the results. At the same time, the masked image contains the spatial information of the optic cup (OC) and the optic disc (OD), which can help the classification network pay more attention to the pathological areas to improve the classification accuracy. Abundant experiments have been performed on the public glaucoma classification dataset, and results show that the proposed method is superior to the existing methods on ORIGA, LAG, RIM-ONE and REFUGE datasets. Furthermore, the network attention experiment also proves the effectiveness of the mask-guided method we proposed.

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Keywords: Glaucoma screening, mask-guided, ROI extraction, mask generation, network attention

Introduction

Glaucoma is an age-wide ophthalmic disease caused by retinal ganglion cell loss and damage to the optic nerve head (ONH). It is already the first irreversible blinding eye disease in the world, and the number of people with glaucoma is expected to reach 118 million worldwide by 2040 [1]. Because of its insidiously nature, the optimal duration of treatment is often missed when glaucoma is diagnosed. Although medical technology at this stage cannot completely eradicate glaucoma, it can help to improve the reliability of glaucoma diagnosis and effectively save patients' vision by early screening. Common glaucoma clinical diagnostic techniques include intraocular pressure (IOP) measurement and ONH examination [2, 3]. These methods are labor-intensive and not suitable for large-scale screening. Therefore, a glaucoma screening program for early and large-scale detection is urgently needed. On the other hand, with the rapid development of computer and artificial intelligence (AI) technology, AI has shown immense potential in screening a series of ophthalmic diseases such as glaucoma, cataract, diabetic retinopathy, etc. By processing fundus images, AI technology can realize automatic glaucoma screening with reliable accuracy and high efficiency, which is suitable for large-scale screening. Currently, there are three main methods of automatic glaucoma screening based on fundus images, which have the following characteristics. The first

is based on image segmentation. These methods rely on image segmentation algorithms to separate key structural regions such as OC and OD, and then calculate clinical indicators based on the segmentation results to assess the risk of glaucoma. The second is based on feature extraction. These methods directly use traditional image processing methods to extract some features from the fundus image, such as texture features, and then train the classifier to diagnose glaucoma directly based on the filtered features. The third is based on deep learning. These methods utilize a classification network to output the risk of glaucoma directly end-to-end. The three main methods still have some difficulties and challenges, which limit their practical application in clinic. The difficulties are summarized as follows:

1) Segmentation accuracy limitation. Segmentation of OC, OD, rim and blood vessels from the fundus image is the guarantee of the accuracy of subsequent screening algorithm. Accurate segmentation algorithm can not only improve the accuracy of the clinical indicators, but also provide a variety of tissue or lesion segmentation results for clinical ophthalmologists, assist doctors in making diagnostic decisions, and greatly improve the efficiency of clinical diagnosis. However, the task of fundus image segmentation is difficult. First, the boundary of OD and OC is badly blurred, which makes it difficult to distinguish. Second, the

target fundus vascular structure to be segmented is staggered and complex, with both large target main vessels and small target capillaries. Third, the resolution of fundus image is low. These problems limit the accuracy of existing glaucoma screening methods based on segmentation.

2) Data limitations. Medical image processing requires a large amount of high-quality data, especially screening algorithms using deep learning techniques. Currently, high-quality collections of fundus image data rely on labels from professional ophthalmologists, so public datasets are scarce. In addition, different physicians depend on different experience for labeling, which can lead to different results, especially for the labeling of OC and OD. Furthermore, because of the differences between acquisition devices, algorithms designed with a single dataset cannot effectively overcome the differences between multiple datasets, resulting in a lower applicability of the algorithm.

3) Poor interpretability. Deep learning algorithms are becoming the mainstream in academia and industry at present, but due to the black-box characteristics of these methods, the interpretability of diagnostic evidence is poor, which is still controversial in practical application.

To improve the limitations above, we propose a glaucoma screening framework based on deep learning method in this paper. We use segmentation method to obtain the corresponding mask image from the color fundus image, which contains the position of OC and OD. However, unlike segmentation-based method, mask images are not directly used for clinical indicators calculation, but serve as an auxiliary input to the glaucoma classification network to guide the network pay more attention to the OC and OD regions of the fundus image. Abundant experiment proves that the method we proposed exactly makes the network focus on the OC and OD area, and finally the accuracy of glaucoma detection is improved. In conclusion, the contributions of this work are as follows:

1. We propose a mask-guided Efficient Net architecture called MG-Efficient Net. In particular, the color fundus images and the

corresponding masks are simultaneously used as the network's input. The color image is the primary source of features to guide the network to classify glaucoma correctly. The mask guides the network's attention to focus on the pathological regions. Validation on several popular public datasets shows that our proposed glaucoma screening method performs better than most other methods.

2. We propose an improved convolutional block Inverted-MB Conv. It is based on the basic structure of MB Conv. The difference is that its depth wise separable convolution layer is moved up and has larger convolution kernels, less batch normalization layer, and less activation function. It has better performance for feature extraction.

3. The network employs a down sampling layer based on dilated convolutions. It abandons the traditional pooling operation and adopts dilated convolution, which increases the receptive field and retains more features in small-scale images, so it has better overall performance.

4. Our method proves that it is feasible for mask to guide network attention, and the accuracy of glaucoma screening is greatly improved. This idea can be helpful to the related work based on deep learning in the future. The remainder of the paper is organized as follows: Section 2 discusses some previous related works. Section 3 describes the proposed method and network architecture in detail. Section 4 is the experimental part, and Section 5 summarizes our work.

Related works

In clinical work, ONH measurement is a commonly used diagnostic method for glaucoma screening. In fundus images, ONH appears as a yellow, bright circular area. Inside it, a smaller circular area in the center is called the OC, and the rest is the neuroretinal rim (NRR).

Ophthalmologists judge whether it is a symptom of glaucoma based on the cup-to-disk ratio (CDR) or ISNT [4] criteria. Figure 1 shows the CDR comparison between normal and glaucoma.

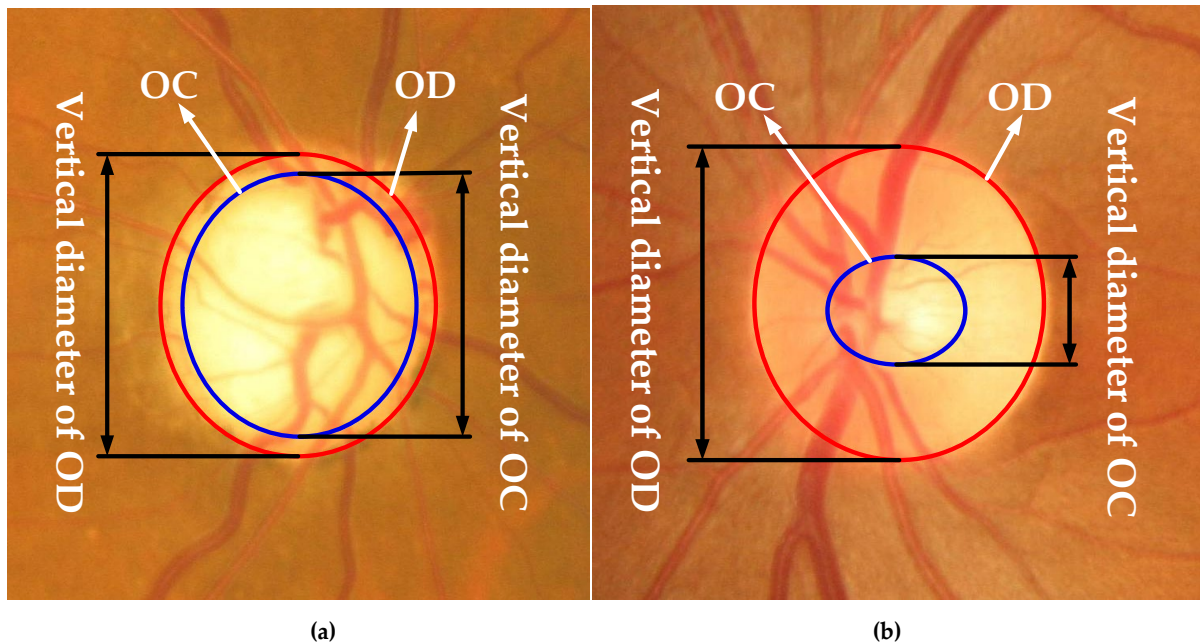


Figure 1: Comparison of CDR. (a) Glaucoma fundus; (b) Normal fundus. The OC is represented with the blue contour and OD is shown with the red contour in both images. The vertical diameter of OC in the glaucomatous images is significantly larger than that in the normal images.

Segmentation of OD and OC

In fundus images, the OD area is usually brighter than the surrounding area, so template matching and pixel classification are often used for OD segmentation. Among the methods based on template matching, Aquino et al. [5] first used morpho-logical transformation and edge detection techniques to process color fundus images. They segmented the approximately circular OD region by Huffman transform. On this basis, Yin et al. proposed a segmentation technique based on a statistical model by fusing the deformable model and applying it to the OC segmentation. Among the methods based on pixel classification, Abramoff et al. [6] proposed a pixel classification method for fundus image segmentation, but how to select pixels for feature ex-traction is a complex problem. Cheng et al. [7] proposed a super-pixel classification method, which obtained the position of each pixel through histogram and center-surround statistics, and then classified it into OD and not OD.

Zilly et al. [8] pro-posed an entropy-based sampling technique for pixel selection to optimize classifier performance. In addition, Mishra et al. first use image enhancement technology to process fundus images, remove blood vessel information and perform image restoration, and finally uses multi-threshold as well as active contour methods for OD and OC segmentation. Most previous work regards OD and OC segmentation as two independent tasks. Zheng et al. introduces preliminary information on geometric interactions such as OD shape, edge thickness, and the relative position, effectively im-proving the segmentation performance. Since deep learning is widely used in computer vision tasks, it has also been introduced into fundus image segmentation. Ronneberger et al. proposed an encoder-decoder convolutional network architecture called U-Net, widely used in image segmentation tasks. Sevastopolsky [9] improved U-Net to make it suitable for OD and OC segmentation tasks.

Shankaranarayana used a fully convolutional network to predict all the pixels of the fundus image and then used an adversarial training technique for segmentation. Fu et al. [10] proposed M-Net, a variant based on the U-Net structure. M-Net can segment OC and OD simultaneously, but since the corresponding labels are independent and no prior information is included, M-Net still treats OC and OD segmentation as two separate problems in essence. Besides, some methods also improve the performance of the segmentation network by introducing new feature information. Jiang et al. [11] proposed an end-to-end deep learning framework named Joint RCNN. It degenerates the original segmentation task into an object detection task.

The two RCNNs extract the boundary information of OD and OC, then use the feature map sharing technology to obtain the spatial constraint information. The attention mechanism improves the network performance and realizes the simultaneous segmentation. Liu et al. [12] proposed a domain adaptation model based on adversarial learning to learn the joint segmentation task of OD and OC. It adopts an adversarial-based structure to match the distribution between the source and target domains to overcome the domain transfer problem. Shankaranarayana et al. [13] proposed a fully convolutional network for the OC and OD segmentation task.

The overall structure of the network is based on U-Net. It designs a multi-modal feature extraction block, which can extract the information of the fundus image and the depth image and then use it as the network's input after fusion. Experiments show that

the additional information introduced by the depth image can effectively improve the accuracy of the OC and OD.

Classification of Glaucoma

One approach to glaucoma classification is first to segment the OD and OC, then extract relevant feature information. However, the quality of segmentation results dramatically affects the accuracy of glaucoma classification, and these methods quickly reach a bottleneck. To address the above issues, Noronha et al. [14] employed Radon transform (RT) to extract higher-order spectral (HOS) cumulates from color fundus images as the input of the classifier, next step used linear discriminant analysis (LDA) to optimize the performance of the classifier. Acharya et al. [15] used Gabor transform with an SVM classifier for glaucoma screening. Mookiah et al. [16] adopted both discrete wavelet change (DWT) and HOS, hereafter adopted a principal component analysis (PCA) method to reduce the acquired features and optimize the classifier performance.

Our previous work [17] extracted the hidden visual features of the Gray Level Co-occurrence Matrix (GLCM) and Gabor Transform from different regions of the fundus image. It adopted Gradient Boosted Decision Tree (GBDT) and ADASYN algorithm for glaucoma classification. Most methods have improved the accuracy of glaucoma screening after introducing hidden features, but how to select hidden features is still a problem worthy of research. Another approach to glaucoma classification is through Deep Neural Networks (DNN). Ref. [18, 19] used deep learning methods to detect glaucoma directly from fundus images but only focused on the contextual information of the OD region while ignoring the global information. Zhao et al. [20] fused convolutional neural networks and random forest regressors for glaucoma screening. This method proposes a regression-based algorithm, which bypasses the intermediate segmentation step and directly estimates the value of the CDR, thus avoiding the external error introduced by the segmentation.

Fu et al. [21] proposed the OD perception ensemble network DE Net for automatic screening of glaucoma. It contains four depth streams for extracting different levels of information on fundus images, optimizing the problem of feature inhomogeneity. Li et al. [22] proposed AG-CNN, an attention-based classification network. Remove redundant information in fundus images by adding attention maps for glaucoma classification. In order to solve the problems in the above methods, such as severely depending on the accuracy of segmentation, difficulty in feature selection, poor interpretability, and so on. This paper presents a mask-guided classification method based on deep learning. Similarly, this method will also segment the fundus image to obtain OD and OC information, but it does not directly calculate clinical indicators, so the influence of segmentation accuracy on the results is very low. Masks can direct the attention of classification networks, resulting in better classification accuracy and more reliable evidence maps.

Proposed Method

The glaucoma screening method we proposed is shown in Figure 2 below. First, we locate the OD and extract the ROI region in the fundus image. Then, the ROI region serves as the input for the segmentation network, which accurately identifies the positions of both the OD and OC. These positional outputs are fused to generate a comprehensive mask image. In the subsequent step, the ROI region is seamlessly integrated with the mask image and fed into the classification network. Following this, we achieve the

glaucoma screening results by processing the amalgamated features and utilizing the classifier's predictions. This holistic

approach ensures a thorough assessment and accurate identification of potential glaucoma cases.

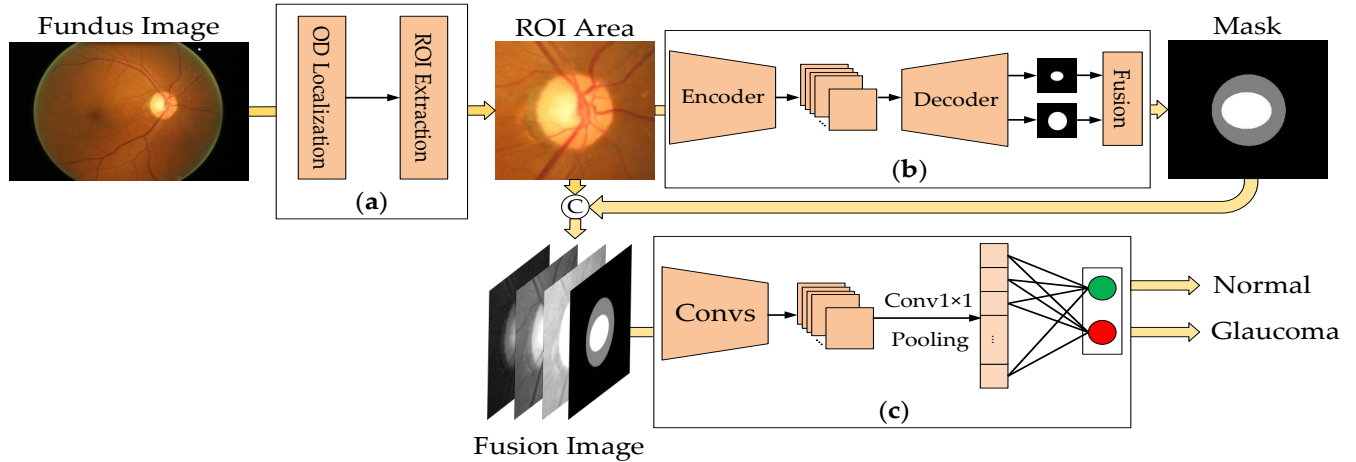


Figure 2: The proposed glaucoma screening method. It consists of three parts: (a) This part includes OD location and ROI extraction, which can segment ROI area from the fundus image; (b) This part is composed of a segmentation network with encoder-decoder structure, which is responsible for generating the corresponding mask according to the ROI area; (c) This part is our proposed glaucoma classification network, called MG-efficient Net, whose input is a fused image of the ROI area and the corresponding mask

ROI Extraction

The datasets used for glaucoma segmentation and classification tasks are high-resolution fundus images captured by specialized medical equipment. Directly using high-resolution fundus images as input to deep neural networks often raises some questions. On the one hand, the computational complexity of the neural network will increase exponentially. As a result, the efficiency of segmentation and classification will drop sharply. On the other hand, since the OD and OC only occupy a small part

of the fundus image, the deep neural network will be easier to over fit, so the results will have extensive bias, which is inevitable and unacceptable for a small amount of data.

To avoid the above problems, we reduce the resolution of the input image by OD positioning and ROI region extraction. Based on our previous work [23], we used the ROI extraction method based on the OD localization, including brightest region extraction, vessel segmentation, and confidence calculation of sliding window.

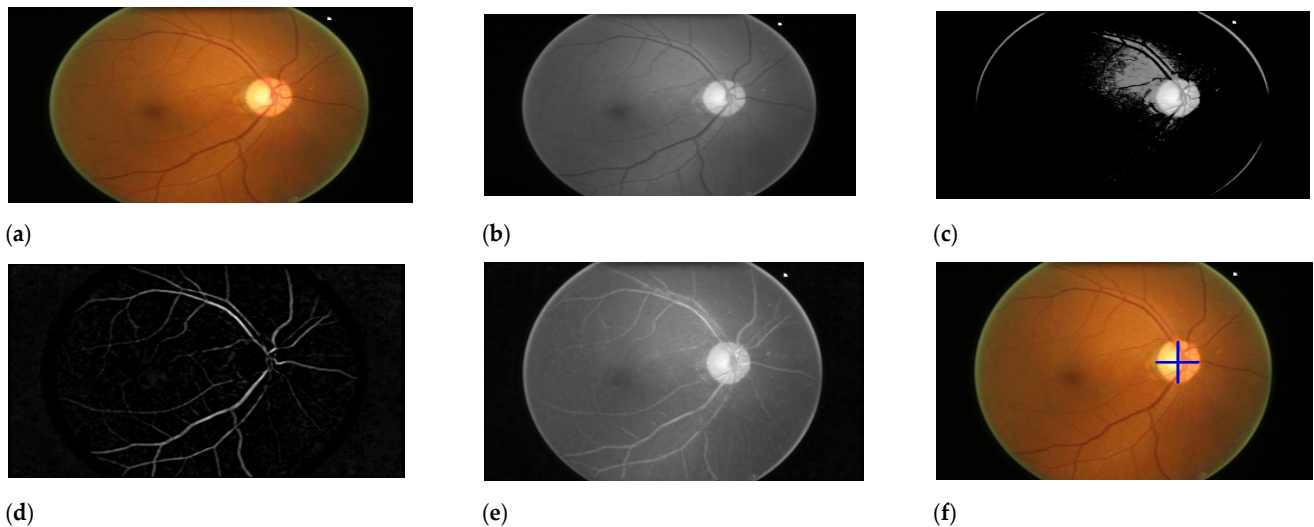


Figure 3: Steps for locating OD. (a) Original color fundus image; (b) Enhanced grayscale fundus image; (c) Brightest area of fundus image; (d) Vascular information; (e) Fusion image; (f) OD localization result. The intersection point of the blue line is the center of the ROI area

Step 1: The fundus image is processed by the morphological transformation (top-bottom-hat) method to obtain an enhanced grayscale map of the fundus image, as shown in Figure 3b. OD is the brightest region in the fundus image, and the brightest part can be extracted by setting the threshold value, as shown in Figure 3c. Step 2: Extracting the vascular information. Firstly, the green channel of the fundus image is processed using contrast-constrained histogram equalization, which can effectively enhance the vascular information in the fundus image. Then, the morpho-logical transformation of the bottom-top-hat is used to

process the fundus image. Finally, the vascular information in the fundus image is extracted by the difference in-information of top-hat transformation and bottom-hat transformation, as shown in Figure 3d. Step 3: Fusion of fundus images and vascular images, as shown in Figure 3e. The average intensity of the local area is calculated by sliding window, and the area with the highest average intensity is considered as the location of the OD, as shown in Figure 3f. Based on the OD localization result, a square area with a width twice the diameter of the OD is extracted as the ROI area. An illustrative example is shown in Figure 4.

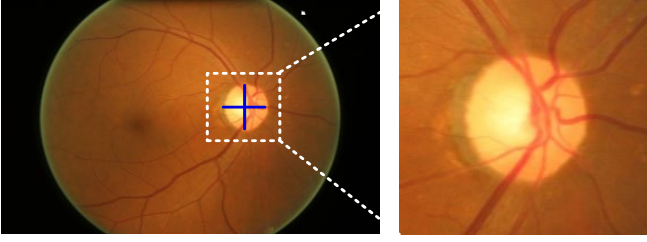


Figure 4: ROI extraction. Taking the intersection of the blue line as the center point and taking twice the diameter of OD as the length, the divided square is considered the ROI area

Mask Generation

In our proposed method, the mask images are obtained by combining the OD and OC segmentation results, which contain three kinds of pixel information. Therefore, the segmentation task needs to be performed first. We use the fully convolutional se-

mantic segmentation network U-Net++ [24], a derivative work of the U-Net network with an encoder-decoder structure, to achieve this goal. Specifically, we use the Res-Net50 Block as encoder and ROI region as input to train the segmentation network.

The segmentation result of OD and OC is a single-channel image with the same size, stored in uint8 type, where 0 represents black pixels and 255 represents white pixels. By using Equation (1), the precision overflow problem in numerical calculation can be effectively avoided, and the mask image can be obtained. The process can be written as:

$$G_m = G_c + \frac{(G_d - G_c)}{2} \quad (1)$$

where G_m is the obtained mask grayscale image, G_d and G_c describe the single-channel segmentation results of OD and OC, respectively. An example of the fusion result is shown in Figure 5d.

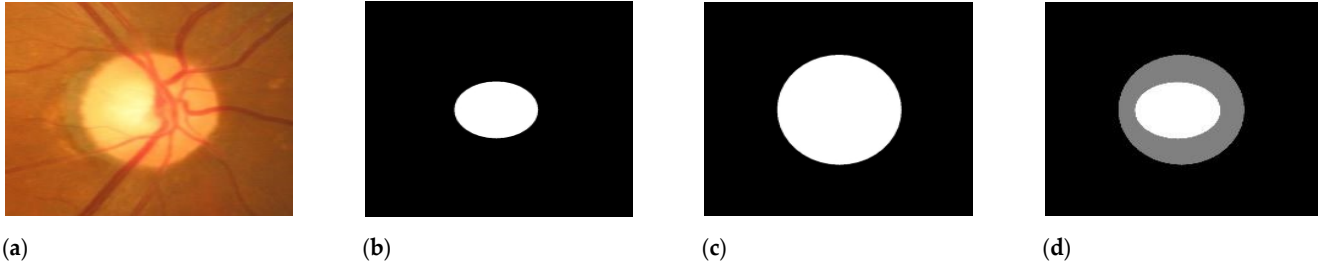


Figure 5: Mask generation process. (a) ROI area; (b) OC segmentation result; (c) OD segmentation result; (d) Mask image.

Glaucoma Classification Network

Efficient Net proposed a robust network structure widely used for classification tasks. Ref. [25] proposed the neural network structure of EfficientNetV2, which has the characteristics of faster training speed and fewer parameters. Inspired by Efficient Net and EfficientNetV2, we propose an improved DNN architecture,

called MG Efficient Net, which aims to improve glaucoma classification through mask images. Figure 6 shows the architecture of the proposed network. The architecture of MG Efficient Net is likely an enhanced version of the original EfficientNetV2, integrating modifications to make it more suitable for the specific task of glaucoma classification using mask images.

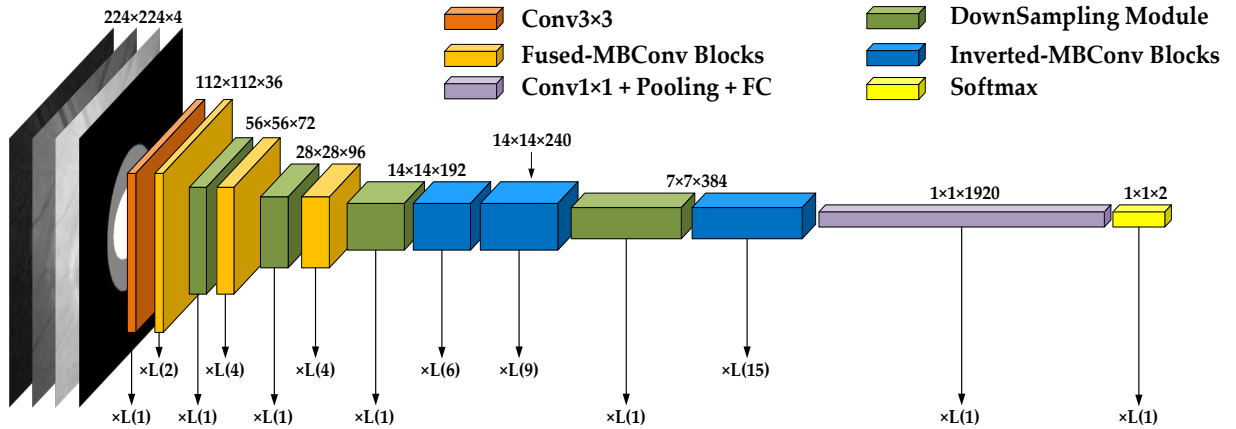


Figure 6: Structure of the proposed MG-efficient Net. Its input is the fusion image of ROI area and mask, so it has more input channels. The symbol $\times L(N)$ represents that the module overlaps N times. The details of Fused-MBConv block, Inverted-MBConv block and Down Sampling module are shown in Figure 7.

Specifically, the network has the following advantages: (i) an EfficientNetV2-S network. Unlike the original network, our proposal introduces extra mask images, including additional information to guide glaucoma classification. Its input is four-channel, which has more channels than most networks. At the same time, to avoid the problem of complex network convergence caused by the increase of information, we expand the number of channels of the convolution layer to 1.5 times the original so that the network can achieve a good balance between accuracy and

parameter quantity. (ii) Inverted-MBConv. The improved convolutional block is our adaptation of MBConv Block, which proved to have better performance on classification task. (iii) Down sampling layer based on dilated convolution. Dilated convolutions are used to avoid pooling operations, thereby retaining more feature information of small-scale images.

EfficientNetV2-S network

The fully convolutional deep neural network EfficientNetV2 has

reliable performance on multi-classification tasks and is widely used. In diagnosing clinical glaucoma, ophthalmologists mainly rely on the CDR and ISNT rules to make diagnostic conclusions. This information exists in the cup-disc area. The original efficient Net structure network is usually used to process color RGB images. The number of input channels is three, and the classification network is not sensitive to the relative position between features. To improve above problems, we add mask images for classification, which contain OD and OC segmentation results, aiming to guide the network to focus on relative position between features, so MG-efficient Net has four input channels. The increased input information may make it difficult for the

network to converge. There-fore, the number of channels of deep convolution is expanded to 1.5 times the original, which improves processing performance of network and convergence speed.

Inverted-MBConv

The main structure of EfficientNetV2 is composed of six groups of convolutional blocks with a different number of channels. Figure 7 illustrates the diverse structures of these convolutional blocks, showcasing the adaptability and scalability that underpins the model's efficiency and effectiveness. This strategic design allows EfficientNetV2 to cater to a wide array of tasks and datasets while maintaining a strong foundation of convolutional architecture.

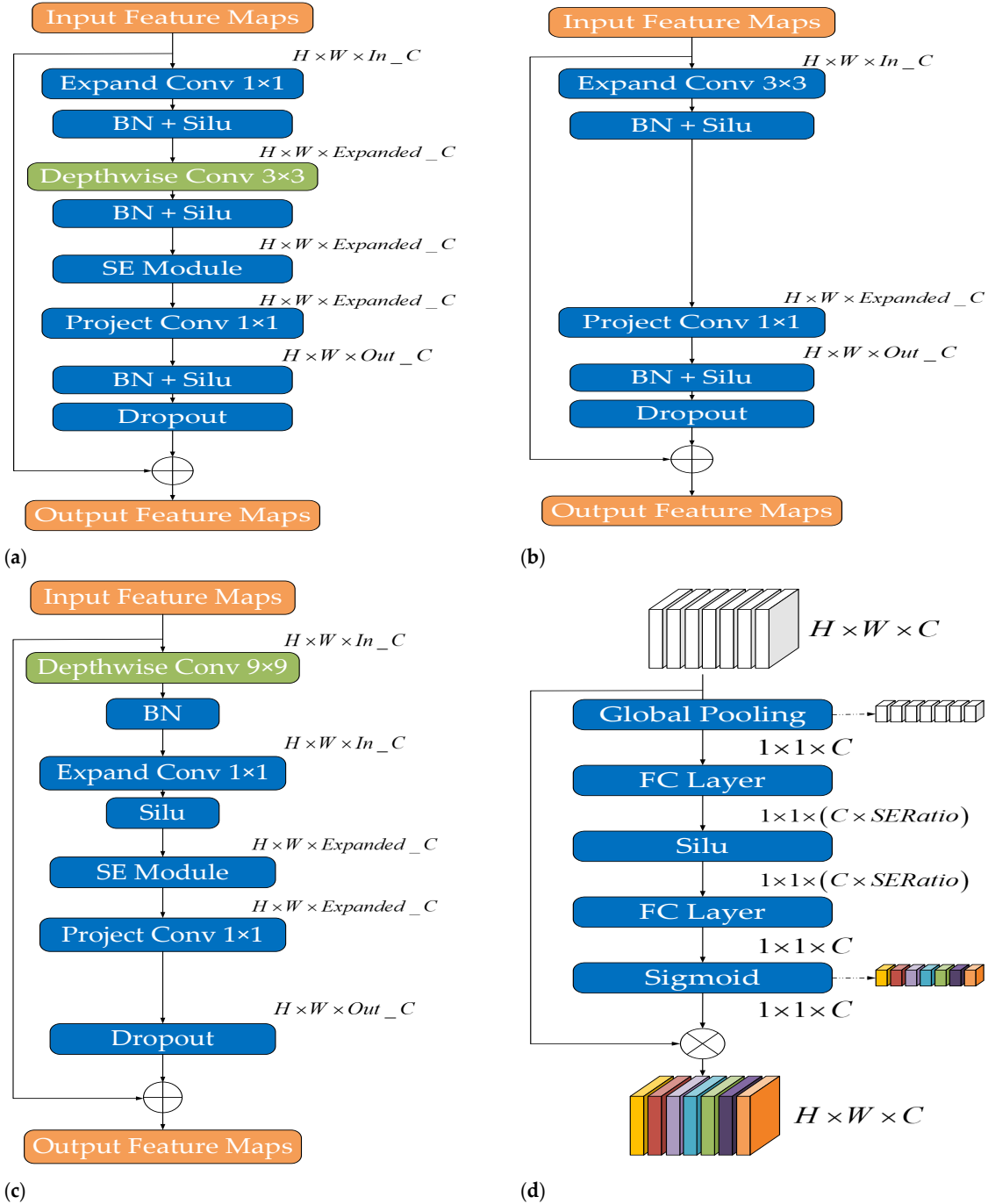


Figure 7: Convolution block structures. (a) Structure of MBConv block; (b) Structure of Fused-MBConv block; (c) Structure of Inverted-MBConv block; (d) Structure of SE module. The formula like $H \times W \times C$ represents the input scale of the nearest block. In particular, In_C and Out_C represent the number of input and output channels for the entire module, respectively. $Expanded_C$ is the product of In_C and expand ratio.

The deep layer of EfficientNetV2 uses MBConv, as shown in Figure 7a, which introduces depthwise separable convolution to reduce the number of parameters significantly and maintain the performance.

The shallow layer uses the Fused-MBConv, as shown in Figure 7b, which incorporates the depthwise separable convolution into the expand convolution, thereby reducing the number of parameters and solving the problem of slow speed. Inspired by ConvNeXt [26], we design the Inverted-MBConv based on

MBConv, as shown in Figure 7c. Precisely, we move up the position of the depthwise separable convolutional layer and increase the convolution kernel, which gives the convolution kernel a larger receptive field and better feature extraction performance.

On the other hand, to cope with the explosion in computation caused by large kernels, we reduce batch normalization and activation functions to achieve a balance between performance and efficiency.

Table 1 Detailed parameters of Fused-MBConv and Inverted-MBConv

Module	NO.	Expand Ratio	Expand Conv	Layers	SE-Ratio
Fused-MBConv	1	1	3×3	2	-
	2	4	3×3	4	-
	3	4	3×3	4	-
Inverted-MBConv	1	4	1×1	6	0.25
	2	6	1×1	9	0.25
	3	6	1×1	15	0.25

Table 1 shows the detailed parameters of Fused-MBConv and Inverted-MBConv in the proposed network. The number of different modules is determined by the order of input to output. The smaller the number, the shallower the module is in the network.

Down sampling layer

Down sampling layer does not use the traditional pooling layer,

but uses dilated convolutions. After the upper module extracts the features, then feature maps are fed to the down sampling layer. First, it performs batch normalization operation. Then ordinary convolution and dilated convolution extract feature, respectively. Finally, the features are compressed into small channels using a convolution with kernel one to pre-vent the depth explosion of the network. Its structure is shown in Figure 8.

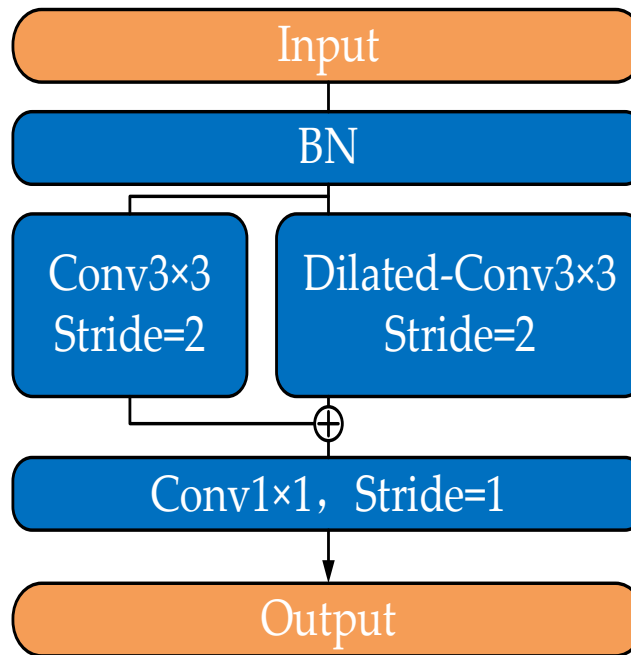


Figure 8: Structure of our proposed down sampling layer.

Since the pooling layer is discarded, more image features are preserved in small image. At the same time, dilated convolution

increases the receptive field without increasing number of parameters, thus the network's overall performance is improved.

Table 2 Parameters of each down sampling layer.

Item	NO.			
	1	2	3	4
Dilated Rate	3	2	2	1
Input Channels	36	72	96	240
Conv Channels	36	72	96	192
Dilated-Conv Channels	72	96	192	192
Output Channels	72	96	192	384

Table 2 shows specific parameter settings of down sampling layers. The number is determined by the order of input to output. The smaller the number, the shallower the module is in the network.

Experiment and results

Dataset introduction

In this paper, five public datasets are used to test the proposed method, including ORIGA [27], LA, RIM-ONE [28], REFUGE [29], and ACRIMA [30]. These datasets all contain high-resolution color fundus images and corresponding glaucoma diagnostic

labels. For example, ORIGA includes 650 sample images, of which 168 are fundus images of glaucoma patients, and the remaining 482 are normal fundus images.

The LAG dataset consists of 2392 diseased images and 3432 normal images. The RIM-ONE dataset contains 344 glaucoma fundus images and 428 normal images. We only used a part of the REFUGE dataset which contained public labels, consisting of 40 glaucoma fundus images and 360 normal images.

The ACRIMA dataset contains 396 glaucoma fundus images and 309 normal images. Table 3 shows a summary of each dataset.

Table 3 Information for each dataset.

Datasets	Number of Images			Applied to	Seg Label Source
	Norm	Glau	Total		
ORIGA	482	168	650	Seg Task & Cls Task	Original
LAG	3432	2392	5824	Cls Task	Generated
RIM-ONE	428	344	772	Cls Task	Generated
REFUGE	360	40	400	Cls Task	Generated
ACRIMA	309	396	705	Cls Task	Generated

In our experiment, we select the ORIGA dataset to train our segmentation network since the ORIGA contains segmentation labels for both OD and OC. The trained segmentation network generates the segmentation results for other datasets. All datasets are used for testing the classification task.

Each original fundus image needs to be extracted from the ROI area and adjusted to a resolution of 224×224 to balance the classification accuracy and training speed.

It involves training a segmentation network using the ORIGA dataset to segment OC and OD regions in fundus images. These segmented images are then used in conjunction with the MG Efficient Net architecture for glaucoma classification.

Training details

For the problem of less training data, we use traditional data enhancement methods to obtain more data samples, including horizontal/vertical flips, rotations, translations, and scale transformations. These methods allow our model to handle more types of input images, improving the accuracy of classification predictions.

In particular, random shearing is also used for the segmentation task of the OC and OD, which further enhances the segmentation network's ability to cope with distorted input images and improves the reliability. The data augmentation scheme is shown in Table 4.

Table 4 Data enhancement scheme

Augmentation technique	Parameter	Applied to
Random vertical flip	Probability: 0.5	Cls Task
Random horizontal flip	Probability: 0.5	Cls Task
Random rotation	Range: [-10, 10]	Cls Task
Random translation	Tuple: (0, 0.1)	Cls Task
Random scaling	Range: [0.9, 1.1]	Cls Task
Random shearing	Tuple: (5.73, 5.73)	Seg Task & Cls Task

All network structures are implemented based on PyTorch and use NVIDIA Ge-Force RTX 3080 to accelerate the training process. The data is divided into a training set and a test set according to the ratio of 8 to 2. The U-Net++ network completes the segmentation task of OD and OC.

Adam optimizer and multiple loss functions that combine the focal loss and the dice loss are used for training, which can solve the problem of unbalanced positive and negative sample data. The classification network uses cross-entropy loss, which can measure minor differences and give the model higher classification accuracy. The loss functions used in the proposed method are:

$$L_{seg} = L_{fl} + L_{dice} \quad (2)$$

$$L_{fl}(p_t) = -\frac{1}{N} \sum_{i=1}^N [\alpha_t (1 - p_t)^\gamma \log(p_t)] \quad (3)$$

$$p_t = \begin{cases} p, & g = 1 \\ 1 - p, & \text{otherwise} \end{cases} \quad (4)$$

$$L_{dice} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{2|p_i g_i|}{|p_i|^2 + |g_i|^2} \quad (5)$$

$$L_{cls} = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)] \quad (6)$$

Here, p_t is the predicted probability for different label values, which in this case OC or OD is 1 and the background is 0. The parameters α_t and γ are used to solve the problem of class imbalance. For the OD and OC area, α_t is set to 0.75, and the rest is set to 0.25. γ set to 2 in our experiment. p_i is the probability that pixel i is predicted to be the target label value, and g_i represents the target true label value. $\hat{y}_i$ is the probability that the classification network predicts a positive class, y_i is the label of the classification, and the positive class is 1 and the negative class is 0.

Evaluation metrics

To compare the proposed glaucoma screening method with other existing DNN-based techniques, we utilize the area under the curve (AUC) of the receiver operating characteristic curve (ROC)

for evaluation. The ROC curve can reflect the sensitivity and the relative relationship between (TPR) and specificity (TNR). Their definitions are shown in Equation (7). In addition, Accuracy is also used as the evaluation index of the classification process, as shown in Equation (8).

$$TPR = \frac{TP}{TP+FN}, TNR = \frac{TN}{TN+FP} \quad (7)$$

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (8)$$

In the above equations, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. AUC ranges from 0 to 1, with larger values representing better classification performance.

Result analysis

In this part, we designed some verification experiments, including the effective-ness of each module, the accuracy of the entire screening framework, and the guidance of the mask-guided method. The results show that each module we designed positively affects the screening accuracy. Besides, the mask-guided method can direct the network to focus on features' relative spatial position and achieve more advanced diagnostic accuracy on most public datasets.

Ablation Study

The proposed method is based on the improved architecture of EfficientNetV2, which includes the following schemes: (a) Increase the input channel to introduce mask; (b) Expand the number of channels to 1.5 times the original; (c) Replace MBConv with Inverted-MBConv; (d) Use improved down sampling layer based on dilated convolutions. We performed ablation experiments to verify the effectiveness of these improvements.

Specifically, the baseline of this experiment is EfficientNetV2-S network, and each network is formed on ORIGA dataset under the same experimental environment. In addition, OD and OC segmentation results are the labels contained originally in the dataset, not the segmentation network generated based on ROI. The kernel size of the Inverted-MBConv Block is 9. This experiment confirms whether they positively affect the accuracy of glaucoma screening. Table 5 describes the experimental results for different network configuration.

Table 5 Results of different network using ORIGA dataset.

NO.	Network	Accuracy	AUC
0	Baseline	0.8124	0.8320
1	Baseline + a	0.8081	0.8243
2	Baseline + a + b	0.8323	0.8492
3	Baseline + a + b + c	0.8404	0.8790
4	Baseline + a + b + c + d (MG-Efficient Net)	0.8474	0.8887

Network 0 has poor glaucoma screening performance on ORIGA datasets with accuracy of 0.8124 and AUC of 0.8320, respectively. After adding the scheme (a), the input of network 1 changes from the original color RGB fundus image to a fused image mix together with the mask. The input channel is also increased from the original three to four. But the accuracy and AUC of the network dropped at the same time. The reason for this problem is that the amount of input information increases while the number of model parameters remain unchanged, thus the network under-fits during the training process, resulting in much lower accuracy and AUC performance. The number of channels and network depth is often two indicators which affect the

performance of deep neural networks. The larger the number of channels, the stronger the ability of network to extract features, and the deeper the network is, the more detailed the extracted features are. However, increasing the number and depth of network channels will also make the network more complex, which will cause too many parameters and make the network difficult to converge. To achieve a balance between performance and complexity, we improve the scheme (b), that is, increase the number of channels of deep convolution by 1.5 times the original. After improvement, the accuracy and AUC of network 2 increased to 0.8323 and 0.8492, respectively.

On this basis, we add the improvement of the scheme (c), replacing the deep MBConv blocks with Inverted-MBConv blocks. The main difference is that the depth-wise conv layer is moved up. This step is first applied to ConvNeXt networks, which reduces the number of complex and inefficient large kernel convolution, and efficient and dense 1×1 convolution is used for feature extraction to reduce the number of network parameters. At the same time, we increase the convolution kernel size of the depth wise conv layer to 9 as compensation for the reduced amount of network parameters. Besides, the improvements also include the reduction of activation functions and normalization layers. This intermediate step resulted in a substantial increase in the accuracy and AUC of the network 3, reaching 0.8404 and 0.8790, respectively.

Finally, we add scheme (d), employing a down sampling layer based on dilated convolutions. This improvement reduces the convolution kernel with stride 2 in the original to 1 and adds a separate down sampling block before it. The introduction of dilated convolution brings some benefits. For example, it does not increase the number of parameters compared to larger convolution kernels and avoids the loss of features compared to pooling operations. Finally, the accuracy and AUC of the network 4 reached 0.8474 and 0.8887, respectively.

Since the performance of network 1 is lower than that of network 0, but the performance of network 2, 3 and 4 is higher than that of the previous one, it is possible that the improvement of scheme (a) will not have a positive effect on the performance of the network. For this reason, we designed additional experiments for further validation. Table 6 shows the results.

Table 6 Results of special network using ORIGA dataset.

NO.	Network	Accuracy	AUC
4	Baseline + a + b + c + d (MG-Efficient Net)	0.8474	0.8887
*	Baseline + b + c + d	0.8293	0.8472

* Special network.

As shown in Table 6, we have designed a special network. Compared with MG-Efficient Net, the special network only uses color fundus image as input. Because of scheme (b), (c) and (d), special network has better performance of convolution layer and down sampling layer. The final results showed that the accuracy and AUC were 0.8293 and 0.8472 respectively.

Compared with baseline, it was improved, but far low-er than the glaucoma screening performance of Network 4. Above experiments show that each module has a positive effect on the overall improvement of the results and can be used simultaneously to obtain the best classification results under a specific configuration of MG-efficient Net. Notably, in the proposed Inverted-MBConv, the size of the convolution kernel is

a vital hyper parameter, and the convolution kernels of different sizes have different abilities to extract deep features. To study the effect of kernel size on glaucoma classification, we try to set it to 3, 5, 7, 9, and 11. The comparative experiment based on network 4 uses the ORIGA dataset, and the results are shown in Figure 9.

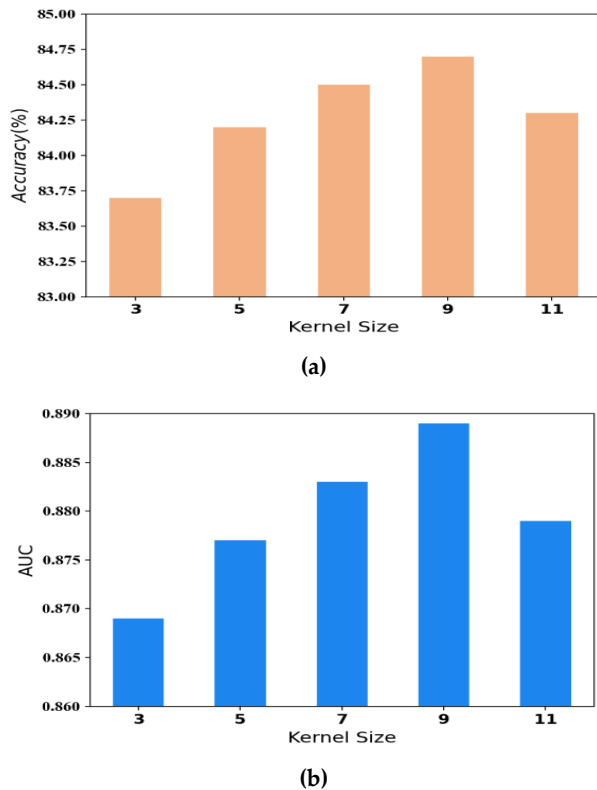


Figure 9: Results of Accuracy and AUC under different kernel sizes using ORIGA dataset. (a) Accuracy result; (b) AUC result.

According to the experimental results, the benefits of using large

convolution kernels are obvious. The accuracy and AUC of glaucoma classification also gradually increase with the increase of the convolution kernel and reach saturation when the kernel size is 9. After that, when the kernel size is further increased to 11, the classification performance of the network drops sharply. Therefore, in the proposed MG-efficient Net network, we adopt the Inverted-MBConv Block with a convolution kernel set to be 9.

Comparison with other existing methods

We compare the performance of the proposed MG-efficient Net with other DNN architectures for glaucoma classification networks. Zhao et al. adopted a semi-supervised learning scheme. The proposed MFPPNet skipped the intermediate step of the OC and OD segmentation, then directly performed CDR regression analysis based on the optic head through a deep learning method. Sreng et al. [31] adopted the method of Ensemble Learning, whose design idea is derived from the whole screening process of the hospital, combining multiple prediction results into a comprehensive prediction, which reduces the generalization error. Li et al. [32] introduced an attention mechanism in deep neural networks and proposed AG-CNN for glaucoma detection, solving the data redundancy problem. Liu et al. offered a novel glaucoma screening and diagnosis network. After the input image is passed through the decoding and classification networks, the segmentation map and classification probability are generated. Then the adversarial network is used to solve the domain shift between the source domain and target domain images in the output space, which improves the accuracy of the network for glaucoma screening. Zhao et al. [33] proposed a multi-task collaborative learning framework MCL-Net. The main structure is a dual-branch neural network, and feature sharing is performed through a feature interaction module to improve network accuracy. Similarly, Hervella et al. [34] proposed a multi-task glaucoma screening framework that fuses fundus image segmentation and classification, leveraging pixel-level and image-level labels during training to improve overall performance.

Table 7 Performance comparison with other methods

Dataset	Methods	Accuracy	AUC
ORIGA	Zhao et al. [20]	-	0.88
	Liao et al. [35]	-	0.88
	Sreng et al. [31]	0.8359	0.8886
	Elangovan et al. [36]	0.8050	-
	Proposed Method	0.8474	0.8887
LAG	Li et al. [37]	0.897	0.960
	Li et al. [32]	0.953	0.975
	Zhao et al. [33]	0.962	0.979
	Elangovan et al. [36]	0.9712	-
	Proposed Method	0.9748	0.9935
RIM-ONE	Diaz-Pinto et al. [30]	0.7121	0.8575
	Li et al. [32]	0.852	0.916
	Zhao et al. [33]	0.824	0.803
	Elangovan et al. [36]	0.8780	-
	Proposed Method	0.9124	0.9613
REFUGE	Sreng et al. [31]	0.9559	0.9510
	Liu et al. [12]	-	0.9715
	Hervella et al. [34]	-	0.976
	Proposed Method	0.9636	0.9880
ACRIMA	Sreng et al. [31]	0.9953	0.9998
	Elangovan et al. [36]	0.9850	-
	Islam et al. [38]	0.9905	0.9891
	Proposed Method	0.9951	0.9998

Table 7 shows the results of comparative experiments on popular public datasets, some of which refer to published studies. Accuracy and AUC are usually the two most important indicators of glaucoma classification [34-36]. Fortunately, our proposed glaucoma screening method achieves the best performance of accuracy and AUC in ORIGA, LAG, RIM-ONE and REFUGE datasets. At the same time, AUC optimal and accuracy suboptimal are also achieved on the ACRIMA dataset.

Due to the differences in the acquisition devices, there are considerable differences among the five datasets we use, which also fully demonstrates the robustness of the method we proposed. In addition, ORIGA, RIM-ONE, REFUGE and ACRIMA datasets belong to mini-datasets, and their samples are all less than 1000, thus proving that our glaucoma screening method also performs well on small datasets. For these reasons, we believe that the proposed mask-guided classification method is of great value for future related research [37, 38].

Comparison of network attention

To verify the effectiveness of proposed mask-guided method, we have carried out a comparative experiment of class activation map. First, we use the EfficientNetV2-S model and our proposed MG-efficient Net model to generate class activation maps of the fundus images using LAG dataset. Class activation map can visually display weight distribution and transfer information during model training.

Concisely, it shows the critical parts that the model found. In the glaucoma classification task, the class activation map shows the image locations on which the model makes classification judgments. In the class activation map, red represents where the model is concerned. When the red color is mainly concentrated in the OC and OD area, it proves that the model has the correct network attention. This makes the classification accuracy of the model higher and the interpretability of the diagnostic results better. The experimental results are shown in Figure 10.

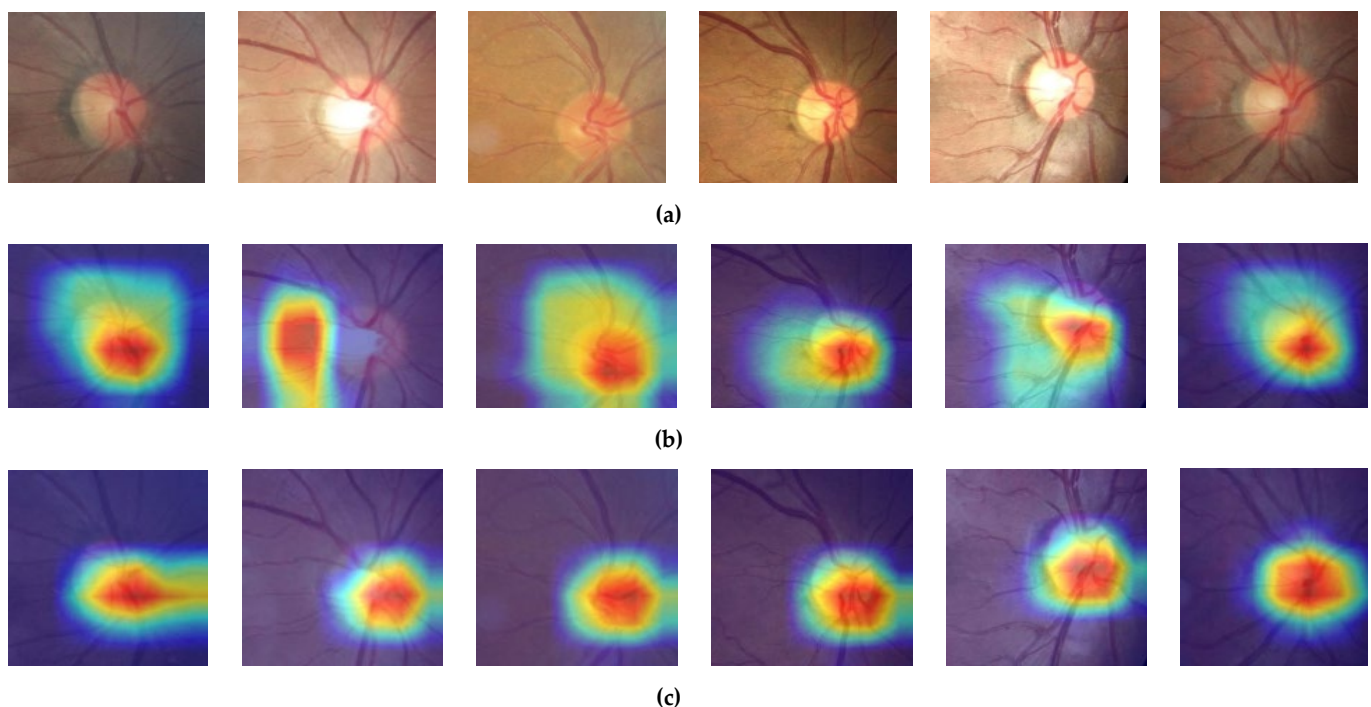


Figure 10: Comparison of network attention. (a) ROI area; (b) Class activation map generated by the EfficientNetV2-S network; (c) Class activation map generated by our proposed MG-efficient Net network.

From the Figure 10, we can observe that compared to EfficientNetV2, the attention of MG-efficient Net is more focused on the OD and OC regions. It should be noted that these areas are usually the essential sources of glaucoma diagnostic basis, which suggests that the proposed MG-efficient Net can focus on the pathological area and its lesions correctly. This enables the MG-efficient Net we improved to have better classification accuracy and better interpretability of the classification results. This also fully demonstrates the effectiveness of the mask-guided method we proposed.

Conclusions

In this paper, we propose a mask-guided method for glaucoma screening based on deep learning. First, after clipping, segmentation and fusion steps, mask images are obtained from fundus images. Then, color fundus images and masks are simultaneously input into our well-designed MG-efficient Net for glaucoma classification. The mask generation process uses

segmentation technology, but our method skillfully reduces the impact of segmentation accuracy on the classification results. At the same time, MG-efficient Net has been carefully designed, including the overall network structure, convolution module and down sampling module, so that it has better classification accuracy. Experiments have shown that our glaucoma screening method achieves optimal performance on ORIGA, LAG, RIM-ONE and REFUGE datasets, and the model is more robust and explanatory. In the next stage, we will further improve the classification accuracy and explore whether the product of the classification task can help to improve the accuracy of the segmentation task.

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Declaration of Conflicting Interests

The author declares that there is no potential conflict of interest for the research and publication of this article.

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